In [7]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
data=pd.read_csv(r"C:\Users\cttcv\Downloads\iris (1)\iris.data",header=None)
```

In [2]:
```
data
```

Out[2]:

	0	1	2	3	4
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Iris-virginica
146	6.3	2.5	5.0	1.9	Iris-virginica
147	6.5	3.0	5.2	2.0	Iris-virginica
148	6.2	3.4	5.4	2.3	Iris-virginica
149	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 5 columns

In [3]:
```
data.columns=['SL','SW','PL','PW','CLASS']
```

In [4]:
```
data
```

Out[4]:

	SL	SW	PL	PW	CLASS
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Iris-virginica
146	6.3	2.5	5.0	1.9	Iris-virginica
147	6.5	3.0	5.2	2.0	Iris-virginica
148	6.2	3.4	5.4	2.3	Iris-virginica
149	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 5 columns

In [5]:
```
data.isnull().sum()
```

Out[5]:
```
SL       0
SW       0
PL       0
PW       0
CLASS    0
dtype: int64
```

In [5]:
```
ip=data.drop(['CLASS'],axis=1)
```

In [1]:
```
import warnings
warnings.filterwarnings('ignore')
```

In [6]:
```
from sklearn import cluster
km=cluster.KMeans(n_clusters=3)
```

```
km.fit(ip)
```

Out[6]:
```
KMeans(n_clusters=3)
```

In [7]:
```
k=km.predict(ip)
print(k)
```

```
[0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0 0 0 0 0 0 2 2 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
 2 2 2 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 1 2 1 1 1 1 2 1 1 1 1
 1 1 2 2 1 1 1 1 2 1 2 1 2 1 1 2 2 1 1 1 1 2 1 1 1 2 1 1 1 2 1 1 1 2 1
 1 2]
```

In [8]:
```
data['predict']=k
```

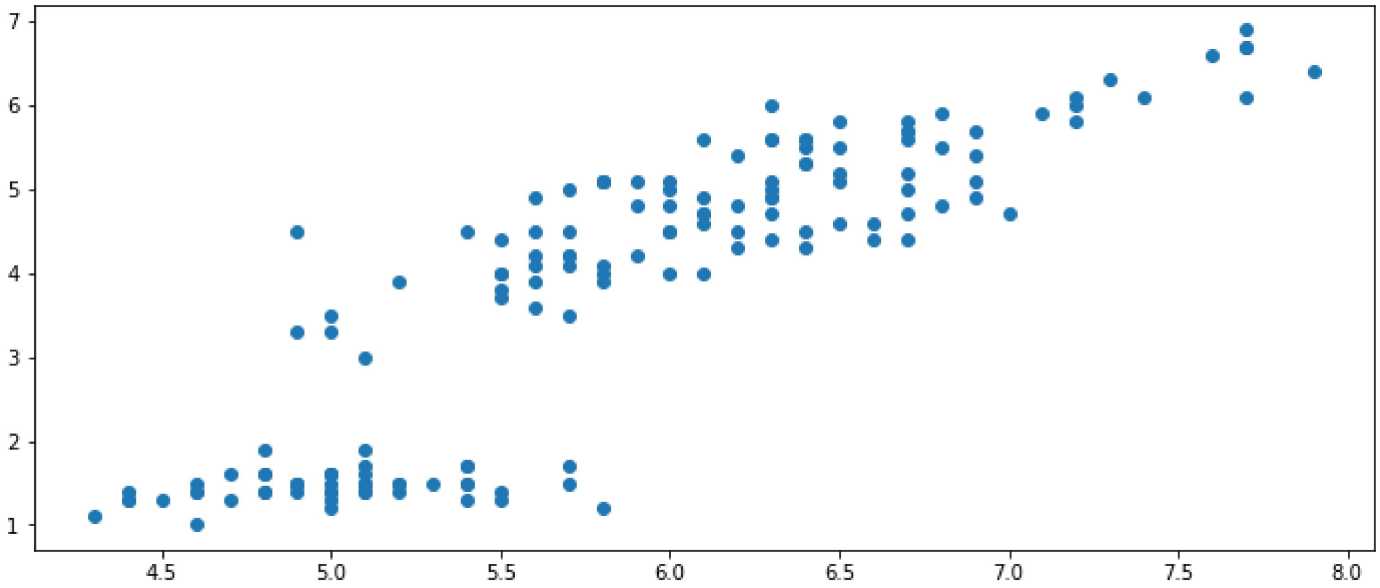
In [9]:
```
data
```

Out[9]:

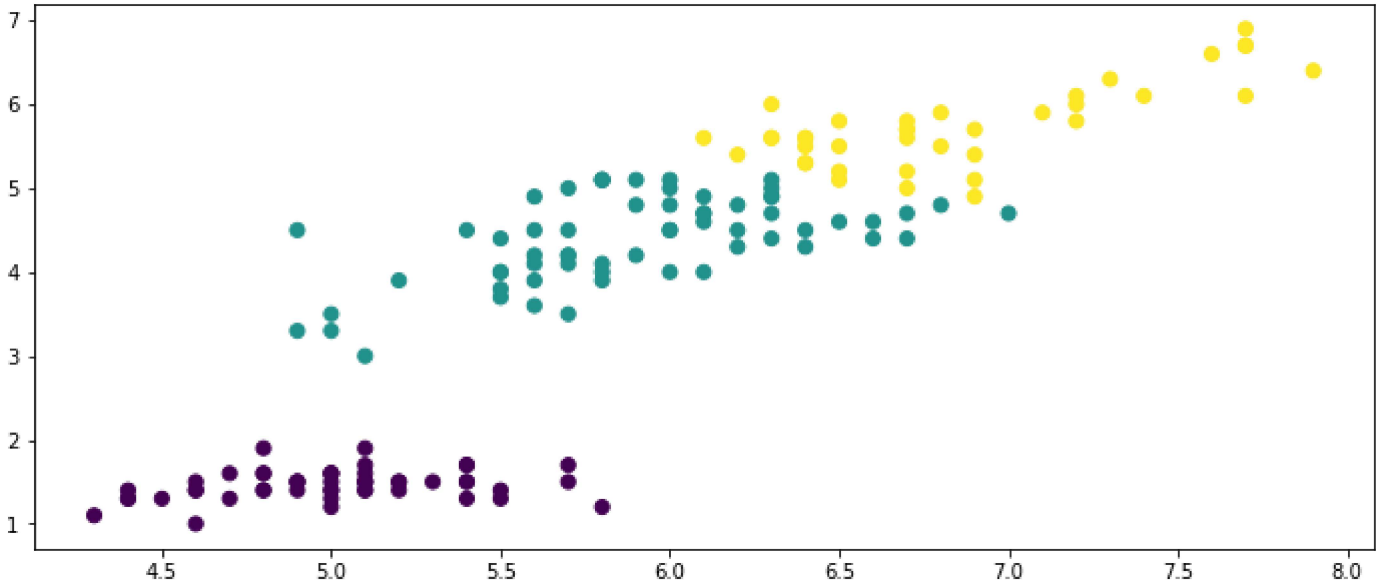
	SL	SW	PL	PW	Class	predict
0	5.1	3.5	1.4	0.2	Iris-setosa	0
1	4.9	3.0	1.4	0.2	Iris-setosa	0
2	4.7	3.2	1.3	0.2	Iris-setosa	0
3	4.6	3.1	1.5	0.2	Iris-setosa	0
4	5.0	3.6	1.4	0.2	Iris-setosa	0
...	...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Iris-virginica	1
146	6.3	2.5	5.0	1.9	Iris-virginica	2
147	6.5	3.0	5.2	2.0	Iris-virginica	1
148	6.2	3.4	5.4	2.3	Iris-virginica	1
149	5.9	3.0	5.1	1.8	Iris-virginica	2

150 rows × 6 columns

In [9]:
```
plt.figure(figsize=(12,5))
plt.scatter(data.SL,data.PL)
plt.show()
```



In [10]:
```
plt.figure(figsize=(12,5))
plt.scatter(data.SL,data.PL,c=k,s=50,cmap='viridis')
plt.show()
```



In [11]:
```
centroid=km.cluster_centers_
print(centroid)
```

```
[[5.006      3.418      1.464      0.244     ]
 [5.9016129  2.7483871  4.39354839 1.43387097]
 [6.85       3.07368421 5.74210526 2.07105263]]
```

In [8]:
```
from sklearn.datasets import make_blobs
X,y=make_blobs(n_samples=300,centers=4,cluster_std=0.6,
               random_state=0)
data=pd.DataFrame(X,y)
```

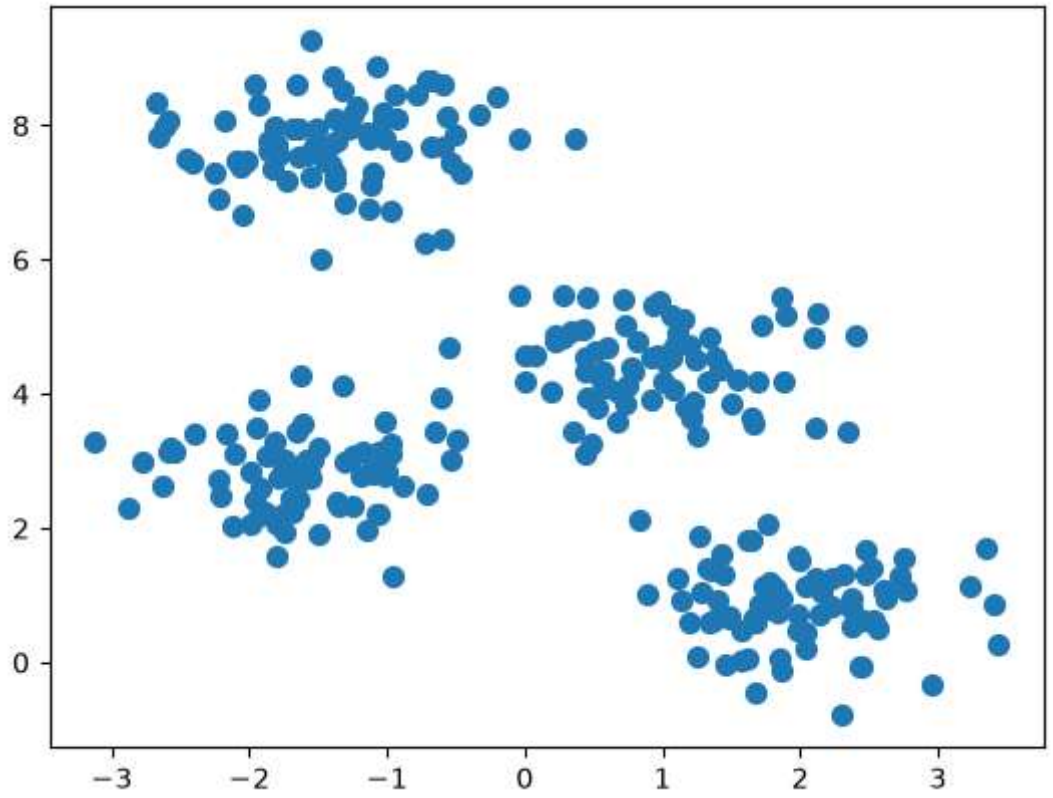
In [9]:
```
data
```

Out[9]:

	0	1
1	0.836857	2.136359
3	-1.413658	7.409623
0	1.155213	5.099619
3	-1.018616	7.814915
1	1.271351	1.892542
...	...	...
1	1.973698	1.579798
1	2.518342	1.391766
0	0.438990	4.535929
3	0.369479	7.791105
2	-1.791458	2.749669

300 rows × 2 columns

In [10]:
```
plt.scatter(X[:,0],X[:,1],s=50)
plt.show()
```



In [11]:
```
from sklearn import cluster

km=cluster.KMeans(n_clusters=4)
km.fit(X)
```

```
C:\anaconda3\Lib\site-packages\joblib\externals\loky\backend\context.py:136: UserWarning: Could not find the number o
f physical cores for the following reason:
[WinError 2] The system cannot find the file specified
Returning the number of logical cores instead. You can silence this warning by setting LOKY_MAX_CPU_COUNT to the numb
er of cores you want to use.
  warnings.warn(
  File "C:\anaconda3\Lib\site-packages\joblib\externals\loky\backend\context.py", line 257, in _count_physical_cores
    cpu_info = subprocess.run(
               ^^^^^^^^^^^^^^^
  File "C:\anaconda3\Lib\subprocess.py", line 548, in run
    with Popen(*popenargs, **kwargs) as process:
         ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "C:\anaconda3\Lib\subprocess.py", line 1026, in __init__
    self._execute_child(args, executable, preexec_fn, close_fds,
  File "C:\anaconda3\Lib\subprocess.py", line 1538, in _execute_child
    hp, ht, pid, tid = _winapi.CreateProcess(executable, args,
                       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
C:\Users\cttcv\AppData\Roaming\Python\Python312\site-packages\sklearn\cluster\_kmeans.py:1425: UserWarning: KMeans is
known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it
by setting the environment variable OMP_NUM_THREADS=2.
  warnings.warn(
```

Out[11]:

▾ KMeans ⓘ ?

▸ Parameters

In [12]:
```
yp=km.predict(X)
data['predict']=yp
```

In [13]:
```
data
```

Out[13]:

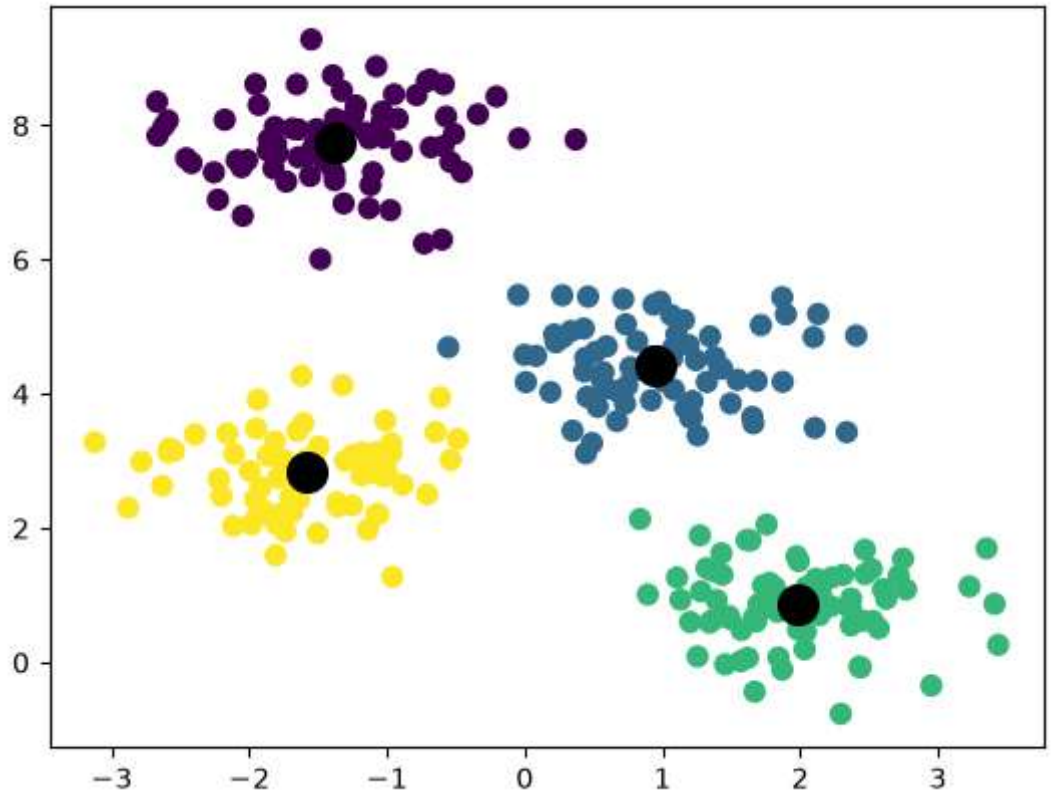
	0	1	predict
1	0.836857	2.136359	2
3	-1.413658	7.409623	0
0	1.155213	5.099619	1
3	-1.018616	7.814915	0
1	1.271351	1.892542	2
...	...	...	...
1	1.973698	1.579798	2
1	2.518342	1.391766	2
0	0.438990	4.535929	1
3	0.369479	7.791105	0
2	-1.791458	2.749669	3

300 rows × 3 columns

In [14]:
```
centroid=km.cluster_centers_
print(centroid)
```

```
[[-1.37324398  7.75368871]
 [ 0.94973532  4.41906906]
 [ 1.98258281  0.86771314]
 [-1.58438467  2.83081263]]
```

In [15]:
```
plt.scatter(X[:,0],X[:,1],c=yp,s=50,cmap='viridis')
plt.scatter(centroid[:,0],centroid[:,1],c='black',s=200)
plt.show()
```



In [16]:
```
def ClusterIndicesNumpy(clustNum,labels_array):
    return np.where(labels_array==clustNum)

ClusterIndicesNumpy(2,km.labels_)
```

Out[16]:
```
(array([  0,   4,   5,  14,  17,  20,  21,  26,  39,  49,  51,  53,  54,
         59,  67,  78,  80,  82,  83,  86,  91,  96,  97, 101, 103, 105,
        106, 112, 114, 117, 118, 121, 123, 124, 125, 126, 128, 133, 154,
        158, 163, 168, 172, 176, 177, 180, 188, 202, 217, 218, 221, 222,
        224, 233, 236, 240, 242, 252, 257, 258, 263, 269, 270, 274, 275,
        276, 280, 281, 283, 284, 285, 290, 291, 295, 296], dtype=int64),)
```

In [17]:
```
X[ClusterIndicesNumpy(2,km.labels_)]
```

Out[17]:
```
array([[ 0.83685684,  2.13635938],
       [ 1.27135141,  1.89254207],
       [ 3.43761754,  0.26165417],
       [ 1.45513831, -0.029199  ],
       [ 2.56936589,  0.5070483 ],
       [ 2.71506328,  1.2908219 ],
       [ 1.77710994,  1.18655254],
       [ 1.6974791 ,  0.86612328],
       [ 1.66382237,  0.66309163],
       [ 1.98894313,  1.50976729],
       [ 2.45509737,  0.62123179],
       [ 1.69687788,  0.75491062],
       [ 1.75644805,  2.05538289],
       [ 1.39263752,  0.92896271],
       [ 2.36960214,  0.95071691],
       [ 2.14043942,  0.70606661],
       [ 1.25471244,  0.08963316],
       [ 1.27955338,  1.05789418],
       [ 1.72955064,  1.14729369],
       [ 2.36923352,  0.79473586],
       [ 1.78194802,  0.90815116],
       [ 1.36155806,  1.36638252],
       [ 2.60137487,  1.08799459],
       [ 1.66909648, -0.43637823],
       [ 2.36790645,  0.55219088],
       [ 2.22707373,  1.26296996],
       [ 2.43934644, -0.07250997],
       [ 1.57322172,  0.48393379],
       [ 1.83375842,  0.75403615],
       [ 1.10550448,  1.26389129],
       [ 1.61152972,  1.82347242],
       [ 1.67841499,  0.60904739],
       [ 1.34195197,  0.59357385],
       [ 1.4281155 ,  1.62435083],
       [ 2.04505527,  1.1251547 ],
       [ 1.67280531,  0.65930057],
       [ 1.13078931,  0.93562086],
       [ 2.43040639, -0.06357093],
       [ 1.56724897,  0.01780906],
       [ 1.86873582,  0.95610376],
       [ 1.64003761,  1.81948989],
       [ 1.43472182,  1.30662037],
```

```
       [ 1.48170052,  0.69007459],
       [ 2.46915842,  1.6787714 ],
       [ 2.31102276,  1.30380848],
       [ 2.62492001,  0.9501944 ],
       [ 2.47019077,  1.31451315],
       [ 1.86985974, -0.10793862],
       [ 1.98436909,  0.48955674],
       [ 2.95195825, -0.34432736],
       [ 2.5270643 ,  0.6178122 ],
       [ 2.04067185,  0.45484511],
       [ 1.61990909,  0.06764529],
       [ 0.89349964,  1.01093082],
       [ 1.97553917,  0.71898913],
       [ 2.7680854 ,  1.08782923],
       [ 3.41085289,  0.87230937],
       [ 1.48859977,  0.65163384],
       [ 2.03169783,  0.19680756],
       [ 2.14917144,  1.03697228],
       [ 1.36678633,  0.63497163],
       [ 2.1139025 ,  1.24743587],
       [ 1.20083098,  0.60167173],
       [ 3.35320909,  1.69958043],
       [ 1.84287117,  0.07269288],
       [ 1.32000621,  1.40428145],
       [ 2.74666646,  1.54543482],
       [ 2.60778282,  1.08890025],
       [ 2.35151259,  0.8280013 ],
       [ 2.22322228,  0.83877343],
       [ 3.22881491,  1.13171965],
       [ 2.29469533, -0.76589199],
       [ 1.8155981 ,  1.11969719],
       [ 1.9736977 ,  1.57979848],
       [ 2.51834185,  1.39176615]])
```

In [18]:
```
from sklearn import cluster

km=cluster.FCM(n_clusters=4)
km.fit(X)
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[18], line 3
      1 from sklearn import cluster
----> 3 km=cluster.FCM(n_clusters=4)
      4 km.fit(X)

AttributeError: module 'sklearn.cluster' has no attribute 'FCM'
```

In [19]:
```
!pip install fuzzy-c-means
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: fuzzy-c-means in c:\users\cttcv\appdata\roaming\python\python312\site-packages (2.0.2)
Requirement already satisfied: numpy<2,>=1.21.1 in c:\anaconda3\lib\site-packages (from fuzzy-c-means) (1.26.4)
Requirement already satisfied: tabulate<1,>=0.8.9 in c:\anaconda3\lib\site-packages (from fuzzy-c-means) (0.9.0)
Requirement already satisfied: tqdm<5,>=4.64.1 in c:\anaconda3\lib\site-packages (from fuzzy-c-means) (4.66.5)
Requirement already satisfied: pydantic<3,>=2.6.4 in c:\anaconda3\lib\site-packages (from fuzzy-c-means) (2.8.2)
Requirement already satisfied: typer<0.10,>=0.9.0 in c:\users\cttcv\appdata\roaming\python\python312\site-packages (f
rom fuzzy-c-means) (0.9.4)
Requirement already satisfied: annotated-types>=0.4.0 in c:\anaconda3\lib\site-packages (from pydantic<3,>=2.6.4->fuz
zy-c-means) (0.6.0)
Requirement already satisfied: pydantic-core==2.20.1 in c:\anaconda3\lib\site-packages (from pydantic<3,>=2.6.4->fuzz
y-c-means) (2.20.1)
Requirement already satisfied: typing-extensions>=4.6.1 in c:\users\cttcv\appdata\roaming\python\python312\site-packa
ges (from pydantic<3,>=2.6.4->fuzzy-c-means) (4.15.0)
Requirement already satisfied: colorama in c:\anaconda3\lib\site-packages (from tqdm<5,>=4.64.1->fuzzy-c-means) (0.4.
6)
Requirement already satisfied: click<9.0.0,>=7.1.1 in c:\anaconda3\lib\site-packages (from typer<0.10,>=0.9.0->fuzzy-
c-means) (8.1.7)
```

In [20]:
```
from fcmeans import FCM
```

In [21]:
```
fm=FCM(n_clusters=4)
fm.fit(X)
```

In [22]:
```
yp1=fm.predict(X)
```

In [23]:
```
data['predict1']=yp1
```

In [24]:
```
data
```

Out[24]:

	0	1	predict	predict1
1	0.836857	2.136359	2	0
3	-1.413658	7.409623	0	2
0	1.155213	5.099619	1	3
3	-1.018616	7.814915	0	2
1	1.271351	1.892542	2	0
...	...	...	...	...
1	1.973698	1.579798	2	0
1	2.518342	1.391766	2	0
0	0.438990	4.535929	1	3
3	0.369479	7.791105	0	2
2	-1.791458	2.749669	3	1

300 rows × 4 columns

In [ ]: